

Inflammatory Demyelinating Neuropathies

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Opinion statement

Early and effective treatment of chronic inflammatory demyelinating polyneuropathy (CIDP) is important to minimize axonal degeneration that occurs secondary to demyelination. The disease course is invariably chronic, so long-term treatment is often required, and adverse effects and costs are important considerations in devising a treatment plan. CIDP responds to prednisone, but long-term treatment can result in significant adverse effects. Azathioprine, mycophenolate mofetil, and cyclosporine can be used as steroid-sparing agents and may facilitate more rapid and successful tapering of prednisone. Intravenous immunoglobulin (IVIg) and plasma exchange are also effective in the treatment of CIDP and can be used in patients who are unresponsive to prednisone or develop steroid-related adverse effects. IVIg may also be used as a first-line treatment, but its cost can be a limiting factor. A few uncontrolled studies have suggested that pulsed weekly methylprednisolone is both effective and well tolerated in the long-term treatment of CIDP. Treatments based on rituximab or cyclophosphamide have also been used in resistant disease.

Variants of CIDP have been described on the basis of their association with specific antibodies or immunoglobulins and their response to specific immunomodulatory treatments. Multifocal motor neuropathy with conduction block responds to IVIg in the majority of patients. However, weakness may slowly worsen over time, and some patients become unresponsive. Anecdotal reports suggest that rituximab may be useful in patients who develop progressive disease.

Placebo-controlled trials in anti-myelin-associated glycoprotein neuropathy suggest that rituximab is effective and, with a combination of prednisone and cyclophosphamide, numbness and strength may improve. Other treatments that may be effective include plasma exchange and IVIg. Treatment is generally started with prednisone, IVIg, or plasma exchange. Rituximab and cyclophosphamide are used only in progressive, treatment-resistant disease because of the potential for serious adverse effects.

Introduction

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an immune-mediated sensorimotor neuropathy that can have a relapsing-remitting or progressive course. The immune attack is primarily directed at the myelin sheath of axons, but associated axonal degeneration is invariable. Early diagnosis and treatment are therefore important to minimize axon loss. Impairment of function and disability can occur because of weakness

or sensory ataxia. Diagnosis is based on clinical features, electrophysiologic testing, analysis of cerebrospinal fluid, and (rarely) nerve biopsy. Criteria commonly used for diagnosis include the Inflammatory Neuropathy Cause and Treatment (INCAT) criteria [1] and the diagnostic criteria of the American Academy of Neurology [2], but many patients who do not fulfill these criteria may also respond to treatment.

Table 1. Treatment options for chronic inflammatory demyelinating polyneuropathy (CIDP) and its variants

Disease	First-line treatment	Other treatments
CIDP	Prednisone,* pulsed methylprednisolone, IVIg,* plasma exchange*	Azathioprine, mycophenolate mofetil, cyclosporine, cyclophosphamide
MADSAM neuropathy	Prednisone, IVIg	? Other immunosuppressants
MMN	IVIg*	? Rituximab
Anti-MAG neuropathy	? IVIg, steroids, plasma exchange	Rituximab,* prednisone + cyclophosphamide*

*Based on placebo-controlled studies.
IVIg—intravenous immunoglobulin; MADSAM—multifocal acquired demyelinating sensory and motor; MAG—myelin-associated glycoprotein; MMN—multifocal motor neuropathy.

Since the initial description of CIDP, variations in the classic clinical presentation have been described. These variants have distinctive clinical characteristics related to the presumed site of pathology within the peripheral nerve and association with specific antibodies. CIDP is a non-length-dependent sensorimotor demyelinating disorder that commonly affects the nerve roots and the distal nerve and over time becomes more or less symmetric. A variant called *multifocal acquired demyelinating sensory and motor neuropathy with conduction block* (MADSAM) was described in 1982 [3]. In this variant, sensorimotor involvement affects the distribution of individual nerves, and persistent conduction blocks are seen on nerve conduction studies. Over time, this variant can evolve into a symmetric neuropathy that is indistinguishable from CIDP in some patients. It is therefore likely that CIDP and MADSAM are related diseases. In 1988, a pure motor variant was described, in which conduction blocks were confined to motor axons, without clinical or electrophysiologic sensory involvement. This disorder was termed *multifocal motor neuropathy with conduction block* (MMN) and was associated with anti-GM1 antibodies in some patients. Like MADSAM, MMN results in progressive weakness over time, but with a paucity of sensory symptoms is often confused with motor neuron disease. Since the initial description of MMN, other cases have been described that lack overt conduction block but are otherwise indistinguishable from MMN [4, Class I; 5••, Class IV]. The presence of overt conduction block may not be essential for diagnosis of MMN, and its absence may deprive these patients of potentially effective treatment.

CIDP can also be associated with monoclonal gammopathies of unknown significance (MGUS). IgM-related neuropathies tend to have a distinctive clinical presentation consisting of demyelination predominantly of sensory axons, resulting in sensory ataxia; some of these patients have anti-myelin-associated glycoprotein (anti-MAG) and anti-sulfated glucuronyl paragloboside (anti-SGPG) antibodies. Distally accentuated conduction slowing that translates into an increased terminal

latency index is the electrophysiologic hallmark of these neuropathies. Neuropathies associated with IgA and IgG are largely indistinguishable from idiopathic CIDP.

TREATMENT PRINCIPLES

Therapy for CIDP and its variants has four general goals: 1) to stop progressive nerve damage to facilitate nerve repair mechanisms; 2) to suppress immune response to myelin components in order to induce a remission; 3) to maintain remission once it is induced; and 4) to minimize cost and adverse effects related to therapy.

Immunomodulatory agents are the mainstay of treatment for CIDP and its variants. Even though the precise immunologic mechanisms for these disorders are not clear, their response to specific immunomodulatory treatments suggests distinct immunologic mechanisms. For example, CIDP responds to steroids, but MMN, which is thought to be a related disease, does not (Table 1). The reason why some of these neuropathies respond to certain agents and not to others is not well understood.

Prednisone is effective in treating CIDP, but the long-term treatment that is often necessary results in severe steroid-related adverse effects. Intravenous immunoglobulin (IVIg) and plasma exchange are also effective [6••,7,8, Class I]. Pulsed methylprednisolone may be effective and is better tolerated than daily steroids in the long-term treatment of CIDP [9,10•, Class IV]. Pulsed oral steroids can therefore be a cost-effective first-line treatment. IVIg or plasma exchange also can be used as first-line treatment or in steroid-resistant patients, but their long-term use is expensive, inconvenient, and can be associated with rare but serious adverse effects. Azathioprine, mycophenolate mofetil, and cyclosporine are also used in treatment-resistant patients or as steroid-sparing agents [11,12, Class IV]. Cyclophosphamide-based treatments may be used as a last resort for intractable disease that continues to progress despite all other treatments [13,14, Class IV]. The role of rituximab in CIDP remains unclear [15, Class IV]. CIDP that is associated with IgG and IgA MGUS is also managed similarly and probably has a similar treatment response.

MADSAM neuropathy, like CIDP, responds to treatment with prednisone and IVIg [16,17, Class IV]. This, along with the fact that some patients with MADSAM evolve into classic CIDP, suggests that these are related diseases that share common pathophysiologic mechanisms and may therefore respond to similar treatment regimens.

MMN responds to IVIg in most patients [18, Class I; 19, Class IV]. MMN does not respond to plasma exchange or prednisone; these treatments can worsen weakness in MMN patients [20, Class IV]. In patients with resistant disease or loss of responsiveness to IVIg, other immunosuppressants are generally used, but this use is based on anecdotal reports. The role of rituximab in MMN remains unclear [15,21, Class IV].

The optimal treatment regimen for anti-MAG neuropathy remains unclear. In a double-blind, randomized, placebo-controlled study, cyclophosphamide and prednisone led to modest improvement in strength and sensation but no significant improvement in function [22••, Class I]. Rituximab also showed promise in the treatment of this neuropathy [23•,24,25, Class IV], and a recently concluded placebo-controlled trial provides additional evidence of efficacy (Personal communication, Marinos Dalakas). Some experts prefer an initial trial of relatively benign treatments such as steroids, IVIg, and plasma exchange [26, Class IV], but the response to these treatments tends to be short-lived.

Treatment

- Treatment for CIDP and its variants should be instituted as early in the disease course as possible, when the predominant pathology is segmental demyelination. Delay in treatment can lead to incomplete response, as recovery from the secondary axonal degeneration that commonly occurs is less efficient. Early diagnosis is therefore a prerequisite to effective treatment.
- The mainstay of treatment is immunosuppression or immunomodulation, but it is also necessary to monitor and manage treatment-related adverse effects. Other important components of management may be symptomatic treatment of neuropathic pain, physical therapy, occupational therapy, and referral for orthotic devices.

Pharmacologic treatment

Prednisone

Standard dose	60 mg/d for 1–2 months. Tapering of the dose can be started after maximum possible improvement has occurred and plateaued for 2 to 4 weeks or if severe adverse effects develop [1, Class II].
Contraindications	Prednisone is contraindicated in systemic fungal infections and known hypersensitivity to components. It is relatively contraindicated in patients with diabetes, peptic ulcer disease, osteoporosis, psychoses, or severe psychoneuroses.
Main drug interactions	Concurrent administration of inducers of hepatic enzymes such as barbiturates, phenytoin, or rifampicin may reduce the effects of prednisone. The action of anticoagulants may be reduced by prednisone.
Main side effects	<i>Fluid and electrolyte disturbances:</i> Sodium and fluid retention, congestive heart failure in susceptible patients, hypokalemia, hypertension. <i>Musculoskeletal:</i> Myopathy, osteoporosis, tendon rupture, vertebral compression fractures, aseptic necrosis of femoral and humeral heads, pathologic fracture of long bones. <i>Gastrointestinal:</i> Peptic ulcer disease, pancreatitis, abdominal distention, ulcerative esophagitis, elevation in liver enzymes. Liver enzyme changes are usually small, not associated with any clinical syndrome, and are reversible upon discontinuation. <i>Dermatologic:</i> Impaired wound healing, thin fragile skin, petechiae, ecchymoses, facial erythema, increased sweating, suppressed reactions to skin tests. <i>Metabolic:</i> Negative nitrogen balance due to protein catabolism. <i>Neurologic:</i> Increased intracranial pressure with papilledema (pseudotumor cerebri), convulsions, vertigo, headache. <i>Endocrine:</i> Menstrual irregularities, development of cushingoid state, secondary adrenocortical and pituitary

	unresponsiveness (particularly in times of stress), suppression of growth in children, decreased carbohydrate tolerance, latent diabetes mellitus, increased requirements for insulin or oral hypoglycemic agents in diabetic patients. <i>Ophthalmic</i> : Posterior subcapsular cataracts, increased intra-ocular pressure, glaucoma, exophthalmos.
Special points	The propensity for and severity of adverse effects is related to age and comorbidities. Decision about treatment should be balanced with potential benefit of treatment and amount of disability.
Cost	The average wholesale price for 30 tablets (50 mg) is about \$9.00.

Methylprednisolone

Standard dose	500 mg by mouth once a week for 3 months. The dose is then tapered as tolerated [9,10, Class IV].
Contraindications	Similar to prednisone.
Main drug interactions	Similar to prednisone.
Main side effects	Similar to prednisone. The treatment may be better tolerated by patients, with short-term adverse effects such as irritability, sleeplessness, unpleasant taste, and dyspepsia lasting for 1 to 2 days after each dose. Weight gain, changes in blood pressure, and glucose intolerance may be milder than with daily prednisone.
Special points	The treatment is generally well tolerated. Osteoporosis is likely in older patients, so prophylactic treatment with bisphosphonates, vitamin D, and calcium is recommended.
Cost	Methylprednisolone is available through a compounding pharmacy at a cost of about \$46 per month.

Intravenous immunoglobulin

Standard dose	<i>Short-term therapy</i> : 2 g/kg per month given on 4 consecutive days. <i>Chronic maintenance therapy</i> : 1–2 g/kg per month given on 2 days every 2 weeks or 4 consecutive days every month [6, Class I].
Contraindications	Patients with severe, selective IgA deficiency (serum IgA < 0.05 g/L) who have known anti-IgA antibodies should receive IVIg only with cautionary measures because of the risk of severe, immediate hypersensitivity reactions, including anaphylaxis.
Main drug interactions	Antibodies in IVIg may interfere with response to live viral vaccines (eg, measles, mumps, and rubella). Use of such vaccines should be deferred until about 6 months after IVIg administration.
Main side effects	Increases in creatinine and blood urea nitrogen have been observed following infusion, predominantly with products containing sucrose as a stabilizer. Progression to oliguria and anuria requiring dialysis can occur, although some patients have improved spontaneously following cessation of treatment. Pyrexia, rigors, dyspnea, cyanosis, hypoxemia, bronchospasm, hepatic dysfunction, leukopenia, pancytopenia, tremor, erythema multiforme, epidermolysis, back pain, abdominal pain, pulmonary edema, seizures, hypotension, thrombosis (pulmonary embolism, myocardial infarction, stroke), or transfusion-related acute lung injury may occur. Mild elevation of liver enzymes may be seen but generally is not associated with symptoms of liver dysfunction.
Special points	Administering IVIg 2 days every 2 weeks rather than on 4 consecutive days every month may reduce the possibility of intertreatment deterioration.
Cost	The cost is about \$10,000 per month for the standard dose.

Plasma exchange

Standard dose	<i>Short-term therapy</i> : Five exchanges (3–4 liters each) every other day over 10 days. <i>Chronic maintenance therapy</i> : One exchange (3–4 liters) every week to every other week [7, Class I].
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Contraindications	This procedure poses high risk for patients with hemodynamic instability from any etiology, unstable angina pectoris, or septicemia.
Main drug interactions	Possible changes in the pharmacokinetics of certain drugs.
Main side effects	Adverse effects related to venous line placement and anticoagulation include vagus nerve syndrome, air embolism, bleeding. Effects related to replacement fluids include fever, chills, nausea, vomiting, and diarrhea. Problems with chronic therapy include alteration in hemostasis with thrombotic complications and infection related to indwelling catheters.
Special points	Plasma exchange is typically used in treatment of acute exacerbations and less commonly in maintenance therapy. It can be used to increase the effectiveness of IVIg in patients with intractable disease.
Cost	About \$10,000 per course of treatment.

Azathioprine

Standard dose	2–3 mg/kg per day is the commonly used dose [27, Class IV].
Contraindications	Hypersensitivity to the drug, pregnancy.
Main drug interactions	Patients receiving allopurinol concomitantly should have a dose reduction of azathioprine, to approximately one third to one quarter of the usual dose. A further dose reduction or alternative therapies should be considered for such patients who have low or absent thiopurine methyltransferase (TPMT) activity.
Main side effects	Myelosuppression with anemia, leukopenia, thrombocytopenia, infection. Gastrointestinal side effects include nausea, vomiting, hepatotoxicity, hepatic veno-occlusive disease, and hypersensitivity pancreatitis.
Special points	TPMT genotyping or phenotyping can help identify patients with low or absent TPMT activity who are at increased risk for severe, life-threatening myelosuppression.
Cost	The generic form of azathioprine costs about \$100 per month.

Mycophenolate mofetil

Standard dose	1000 mg by mouth twice a day [12, Class IV].
Contraindications	Hypersensitivity to the drug.
Main drug interactions	Interactions with acyclovir, antacids containing magnesium and aluminum hydroxide, cholestyramine, cyclosporine, ganciclovir, oral contraceptives, sevelamer, trimethoprim/sulfamethoxazole, norfloxacin, metronidazole, rifampin. Live attenuated vaccines may be less effective.
Main side effects	Diarrhea, leukopenia, sepsis, vomiting, higher frequency of infections.
Special points	The onset of action may be quicker than that seen with azathioprine.
Cost	The cost is about \$1000 per month. No generic form is available.

Rituximab

Standard dose	375 mg/m ² by IV infusion once a week for 4 weeks [21,23•, Class IV].
Contraindications	Hypersensitivity to rituximab.
Main drug interactions	None
Main side effects	The most common adverse reactions observed are infusion reactions, fever, chills, infection, asthenia, and lymphopenia. The most important serious adverse reactions include infusion reactions, mucocutaneous toxicities, hepatitis B reactivation with fulminant hepatitis, progressive multifocal leukoencephalopathy (PML), other viral infections, cardiac arrhythmias, renal toxicity, and bowel obstruction and perforation.
Special points	Complete blood count with differential, peripheral CD20 ⁺ cells, and renal function should be monitored. Cardiac monitoring should be done during and after infusion in patients with pre-existing cardiac disease or if arrhythmias develop during or after infusions. Screening for

hepatitis B in high-risk persons may be considered prior to initiation of therapy. Carriers and patients with prior hepatitis B infection should be monitored closely for clinical and laboratory signs of infection during therapy and for up to 1 year following completion of treatment.

Cost The cost of a 4-week course is about \$23,000, based on an average wholesale price of \$582 per 10-mg vial.

Cyclophosphamide

Standard dose	0.75–1.0 g/m ² by IV pulse monthly for a maximum of 6 months unless there is sustained improvement over three courses, in which case treatment could be shorter [13,14, Class IV].
Contraindications	Suppressed bone marrow function, hypersensitivity to the drug.
Main drug interactions	Chronic treatment with barbiturates can increase metabolism and leukopenic activity of cyclophosphamide. The drug can inhibit cholinesterase activity and potentiate action of succinylcholine.
Main side effects	Increased incidence of urinary bladder, lymphoproliferative, and myeloproliferative malignancies. Other effects can include teratogenicity in pregnant women, impairment of fertility, hemorrhagic cystitis, acute cardiac toxicity, infections, nausea, vomiting, and alopecia. Rarely interstitial pneumonitis can occur with prolonged therapy.
Special points	White blood cell (WBC) count should be monitored weekly. The intention is to decrease the WBC count to 1500/mm ³ to 2000/mm ³ at 2 weeks after infusion. The dose of cyclophosphamide can be increased by 25% on subsequent pulses if reduction in WBC count is not achieved.
Cost	The cost of the drug is about \$30 per dose, but the actual cost is much higher when the costs of administration, monitoring, and treatment of possible toxicity are factored in.

Disclosure

No potential conflicts of interest relevant to this article were reported.

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